



**College of Computing and Informatics
Department of Computer Science**

Course Syllabus

Course Title

Introduction to Artificial Intelligence

Course Code
1501330

Fall 2026/2027

Course Syllabus

Course Title	Introduction to Artificial Intelligence				
Course Code	1501330				
Credit Hours	3	Contact Hours	3		
Course Category	Program required				
Course Type	Compulsory				
Pre-requisite(s)	1411215 - Data Structures OR 1501215 - Data Structures OR 1411214 - OR 1501214 - Programming With Data Structures OR 1501220 - Python Programming				
Co-requisite(s)					
Semester	Fall 2026/2027				
Instructor Name	Email	Classroom	Section	Session Type	Time
Muhammad Anwar Shahid	mshahid@sharjah.ac.ae	C12-004D	31	Lecture Undergraduate	MW 1230 - 1345
Isam Mashhour Hasan Al Jawarneh	ijawarneh@sharjah.ac.ae	A8-106	62	Lecture Undergraduate	TR 1230 - 1345
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Course Description:

This course will provide an introduction to the fundamental concepts and techniques in the field of artificial intelligence. Topics covered in the course include basic search, heuristic search, game search, constraint satisfaction, knowledge representation, and machine learning algorithms, Perception, Robotics, and Natural Language Processing.

Course Objectives/Goals:

The goals of this Course are to:

- To have an appreciation for and understanding of both the achievements of AI and the theory underlying those achievements as well as the importance of AI to modern computing studies
- To acquire a solid understanding of the notion of search that play an important role in AI programs
- To understand the importance knowledge representation and powerful automated reasoning
- To have a basic understanding of some of the more advanced topics of AI such as learning, natural language processing, agents and robotics, expert systems, and planning

Course Learning Outcomes:

By the end of successful completion of this Course, the student will be able to:

1. Define rational agents for applications and describe their task environments
2. Analyze problems and express them as search problems
3. Choose and apply appropriate search techniques to solve problems

4. Differentiate between blind, heuristic, CSP, and adversarial search techniques
5. Apply knowledge representation and probabilistic reasoning techniques to solve AI problems
6. Implement machine learning algorithms to solve real-world problems

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	- Define rational agents for applications and describe their task environments - Analyze problems and express them as search problems - Choose and apply appropriate search techniques to solve problems - Differentiate between blind, heuristic, CSP, and adversarial search techniques
Apply computer science theory and software development fundamentals to produce computing-based solutions.	- Apply knowledge representation and probabilistic reasoning techniques to solve AI problems - Implement machine learning algorithms to solve real-world problems
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Choose and apply appropriate search techniques to solve problems - Apply knowledge representation and probabilistic reasoning techniques to solve AI problems - Implement machine learning algorithms to solve real-world problems
Communicate effectively in a variety of professional contexts.	- Differentiate between blind, heuristic, CSP, and adversarial search techniques
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.	- Apply knowledge representation and probabilistic reasoning techniques to solve AI problems - Implement machine learning algorithms to solve real-world problems
Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	- Define rational agents for applications and describe their task environments - Analyze problems and express them as search problems - Choose and apply appropriate search techniques to solve problems - Differentiate between blind, heuristic, CSP, and adversarial search techniques
Design, implement, and evaluate a computing and	

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
multimedia-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Choose and apply appropriate search techniques to solve problems - Apply knowledge representation and probabilistic reasoning techniques to solve AI problems - Implement machine learning algorithms to solve real-world problems
Communicate effectively in a variety of professional contexts.	- Differentiate between blind, heuristic, CSP, and adversarial search techniques
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.	- Apply knowledge representation and probabilistic reasoning techniques to solve AI problems - Implement machine learning algorithms to solve real-world problems
Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based system.	- Apply knowledge representation and probabilistic reasoning techniques to solve AI problems - Implement machine learning algorithms to solve real-world problems
Apply knowledge of mathematics, computer science, computer engineering, and computational biology appropriate to the discipline.	- Apply knowledge representation and probabilistic reasoning techniques to solve AI problems - Implement machine learning algorithms to solve real-world problems
Analyze a problem, identify and define the computing requirements appropriate to its solution.	- Define rational agents for applications and describe their task environments - Analyze problems and express them as search problems - Choose and apply appropriate search techniques to solve problems - Differentiate between blind, heuristic, CSP, and adversarial search techniques
Design and evaluate a software system, process, or program to meet desired needs.	- Choose and apply appropriate search techniques to solve problems - Apply knowledge representation and probabilistic reasoning techniques to solve AI problems - Implement machine learning algorithms to solve real-world problems
Apply design and development principles in the construction of software and informatics- based solutions of varying complexity.	- Implement machine learning algorithms to solve real-world problems

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
Function effectively on teams and communicate effectively with a range of audiences.	- Apply knowledge representation and probabilistic reasoning techniques to solve AI problems - Implement machine learning algorithms to solve real-world problems

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	CLOs
1	Ch 1-2 - Introduction: What Is AI?, The Foundations of AI., The History of AI, The State of the Art - Intelligent Agents: Agents and Environments, Good Behavior: The Concept of Rationality, The Nature of Environments, The Structure of Agents.	- Assignment 1 - Quiz #1	1
2	Ch 3.1-4, 3.5-7 - State Space Search: Problem-Solving, Searching for Solutions, Uninformed Search Strategies - State Space Search: Heuristic Search	- Assignment 2 - Quiz #2	1,2
3	Ch 3.1-4, 3.5-7 - State Space Search: Problem-Solving, Searching for Solutions, Uninformed Search Strategies - State Space Search: Heuristic Search	- Assignment 2 - Quiz #2	1,2,3
4	Ch 4.1-2, 5.1-2, 5.4 - Local Search: Algorithms and Optimization Problems, Continuous Spaces - Adversarial Search:	- Assignment 3 - Quiz #3	2,3
5	Ch 4.1-2, 5.1-2, 5.4 - Local Search: Algorithms and Optimization Problems, Continuous Spaces - Adversarial Search:	- Assignment 3 - Quiz #3	4
6	Ch 5.3, 5.5, 6.1-2 - Adversarial Search: Alpha–Beta, Stochastic Games - Constraint Satisfaction Problems (CSP): Defining Constraint Satisfaction Problems, Constraint Propagation: Inference in CSPs	- Assignment 3 - Quiz #3	4
7	Ch 5.3-4, 7.1-4 - Constraint Satisfaction Problems (CSP), Backtracking Search for CSPs, Local Search for CSPs - Logical Agents: Knowledge-Based Agents, The Wumpus, Logic, Propositional Logic: A Very Simple	- Assignment 3 - Quiz #3	4
8	Ch 7.5-7 Logical Agents: Propositional Theorem, Effective Propositional Model, Agents Based on Propositional Logic	- Assignment 4 - Midterm	4
9	Ch 8.1-2, 8.3-4 - Uncertainty in Artificial Intelligence: Representing Knowledge in an Uncertain. - Introduction to Machine Learning: Supervised and Unsupervised Learning	Assignment 4	3,4
10	Ch 13	Project	3,5

Week	Topic	Comments*	CLOs
	Learning from Examples: Forms of Learning, Decision tree, Evaluating and Choosing the Best		
11	Ch 14.1-5, 18.1-4 - Learning from Examples: The Theory of Learning, Regression and Classification with Linear Models, Artificial Neural Networks, Nonparametric Models - Learning from Examples: Support Vector Machines, Ensemble, Practical Machine Learning .	Project	3,5,6
12	Ch 18.5-8 Perception: Image Formation, Early Image-Processing Operations, Object Recognition by appearance, Reconstructing the 3D World	Project	3,5,6
13	Ch 18.5-8 Perception: Image Formation, Early Image-Processing Operations, Object Recognition by appearance, Reconstructing the 3D World Ch 18.9-12 Robotics: Introduction, Robot Hardware, Robotic Perception, Planning and Move, Application domains.	Project	3,5,6
14	Ch 22.1-3 Natural Language Processing: Language Models, Text Classification, Information Retrieval.	Project	3,5,6
15	Ch 23.1-5 Natural Language for Communication: Phrase Structure Grammars, Syntactic Analysis, Semantic Interpretation, Applications such as Machine Translation, Speech Recognition	Project	3,5,6
16	Final Exam		3,5,6

Information on out-of-class assignments with due dates for submission

Assignment/Activity	Due Date	Comments
Assignment 1	End of week 1	
Assignment 2	End of week 3	
Assignment 3	End of week 7	
Assignment 4	End of week 9	

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Assignments	Weeks 1-15	10
Quizzes	Weeks 1-15	15
Projects	Weeks 10-15	10
Project Presentation	Weeks 14	5
Midterm Exam	Week 8/9	20
Final Exam	Week 16	40
Total		100

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
Define rational agents for applications and describe their task environments	- Lectures - Problem Solving	- Assignments - Exams	70% of student achieve 70% of the CLO	- Critical and analytical thinking - Teamwork or collaboration	
Analyze problems and express them as search problems	- Lectures - Problem Solving	- Assignments - Exams	70% of student achieve 70% of the CLO	- Problem solving	
Choose and apply appropriate search techniques to solve problems	- Case studies - Lectures - Problem Solving	- Assignments - Exams - Projects	80% of student achieve 80% of the CLO	- Critical and analytical thinking	
Differentiate between blind, heuristic, CSP, and adversarial search techniques	- Case studies - Lectures - Presentations	- Assignments - Exams - Projects	80% of student achieve 80% of the CLO	- Critical and analytical thinking	
Apply knowledge representation and probabilistic reasoning techniques to solve AI problems	- Case studies - Lectures - Presentations	- Assignments - Exams - Projects	80% of student achieve 80% of the CLO	- Problem solving	
Implement machine learning algorithms to solve real-world problems	- Case studies - Lectures - Presentations	- Assignments - Exams	70% of student achieve 70% of the CLO	- Oral communication - Teamwork or collaboration	

Scope and Nature of Assessment Tools:**Assignments:**

Students will have 3 assignments as described in the weekly schedule.

Midterm:

This assessment evaluates students' understanding of fundamental Artificial Intelligence concepts covered in the first half of the course. Students will demonstrate their ability to analyze problems, formulate them as search problems, and apply appropriate search techniques. The midterm will assess both theoretical understanding and problem-solving skills in topics such as intelligent agents, uninformed and informed search, adversarial search, and constraint satisfaction problems. The midterm will be graded, and feedback will be provided to support students' progress.

Quizzes:

Students are expected to attend lectures and complete quizzes conducted during class. These quizzes assess students' understanding of recently covered topics, including key AI concepts and problem-solving techniques. Quizzes may include multiple-choice questions, short answers, or simple problem-solving tasks. The quizzes will be graded, and feedback will be shared with students to reinforce learning.

Final Exam:

This assessment evaluates the comprehensive knowledge and skills acquired throughout the course. Students will demonstrate their ability to analyze problems, apply Artificial Intelligence techniques, and reason about solutions across a range of topics. The exam will include both theoretical and problem-solving components, covering areas such as search strategies, knowledge representation, reasoning under uncertainty, machine learning fundamentals, and natural language processing. The final exam will be graded, and feedback will be provided to highlight strengths and areas for improvement.

Project:

Each project team will consist of two to four members. One member will be designated as the project team leader; this person will be responsible for coordinating the activities of the team. The duties of the members of each team will be decided by the team as a whole and need not be equal on each phase of the project, but work load must/should somehow balance up equally towards the end of the semester. Each team will specify, design, and implement, a small AI application. Below is the rubric will be used to grade the project.

In-class Activities/Teaching Methodology:

The course includes lectures, lab practice, assignments, iTBLs (social learning), group projects (report & demo).

in-class work:

Students are divided into teams. After each chapter/lecture, teams are given an exercise to discuss and solve collaboratively. In-class work enhances the problem solving skills, while developing the interpersonal skills.

Lab Sessions:

Some of the classes will be held in the lab to introduce the Programming Language that will be used during the term. The students will be asked to design and implement their projects using Programming language such as R and Python.

Teaching and Learning Resources:

Textbook(s):

Russell, S. & Norvig, P. (2020). Artificial Intelligence, A Modern Approach. 4th ed. Pearson Education Inc.

Recommended Readings:

Rich, E. & Knight, K & Nair, B. (2017). ARTIFICIAL INTELLIGENCE. 3rd ed.

Liu, Y. (2017). Python Machine Learning. - ed. Packt Publishing.

Other Resources:

- Python; Jupyter Notebook.

- Open lab for students to solve their programming assignments

Attendance policy:

Attendance is compulsory. A student missing 10% of the total allocated course hours will receive 1st warning notice and a student missing 15% will receive 2nd warning notice. A student missing 20% will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Notes:

Attendance policy:

Attendance is compulsory. A student missing 10% (6 hours) of the total allocated course hours will receive 1st warning notice and a student missing 15% (9 hours) will receive 2nd warning notice. A student missing 20% (12 hours or more) will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Practical Aspect of the course:

Students are given 5 to 6 programming assignments covering the topics covered in the course. Each assignment is usually made of a number of problems. Usually one of these problems requires a lot programming. The other problems are designed in a way to give the student the chance to implement to a particular data structures to solve the given problem.

In-class Activities/Teaching Methodology:

Lecturing, Q&A on the board, program implementation using IDE, team-based work on problems (peer learning).

Teaching and Learning:

The course includes two lectures per 15 weeks, which include quizzes, and assignments. The quizzes and assignments are on the same level of difficulty with the midterm and final exams. The assignments were designed to enforce the problem solving and critical thinking skills.

Grading rubrics:

Grading Rubric for Course Projects					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Introduction	10%	No clear introduction, absent, or no	The introduction states the main topic but does	The introduction states the main	The introduction is engaging, states the main

Grading Rubric for Course Projects					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
		evidence.	not adequately preview the structure of the report.	topic and previews the structure of the report.	topic and previews the structure of the report.
Data	20%	Data are inadequate and selected following the wrong procedure.	Data are selected but procedure of data selection is not followed.	Data selected following the standard procedure but not described in report.	Data are correctly selected following standard procedure and adequately described in report.
Analysis and Reporting of Results (if applicable)	20%	Calculated wrong measures and analysis not relevant.	Calculated correct measures but analyses are not correct.	Calculated correct measures but analyses are not adequate.	Calculated correct measures and analyses are correct and provide more insights.
Authenticity and Citation*	20-30%	Some contents are paraphrased; plagiarism is most likely.	Some content is paraphrased; plagiarism is likely.	Some content is paraphrased and no suspicion of plagiarism.	All contents are original and well written.
Writing	20-30%	Errors in language that cause the content to be mostly incomprehensible.	Some errors in language that often inhibit comprehension.	Some errors in language that do not often inhibit comprehension.	Almost no error in language and well written.
Teamwork	10%	Team did not collaborate or communicate well.	Team did not collaborate or communicate well. A lack of mutual respect was frequently noted.	Team did collaborate or communicate well.	Team did collaborate or communicate well with high level mutual respect.